

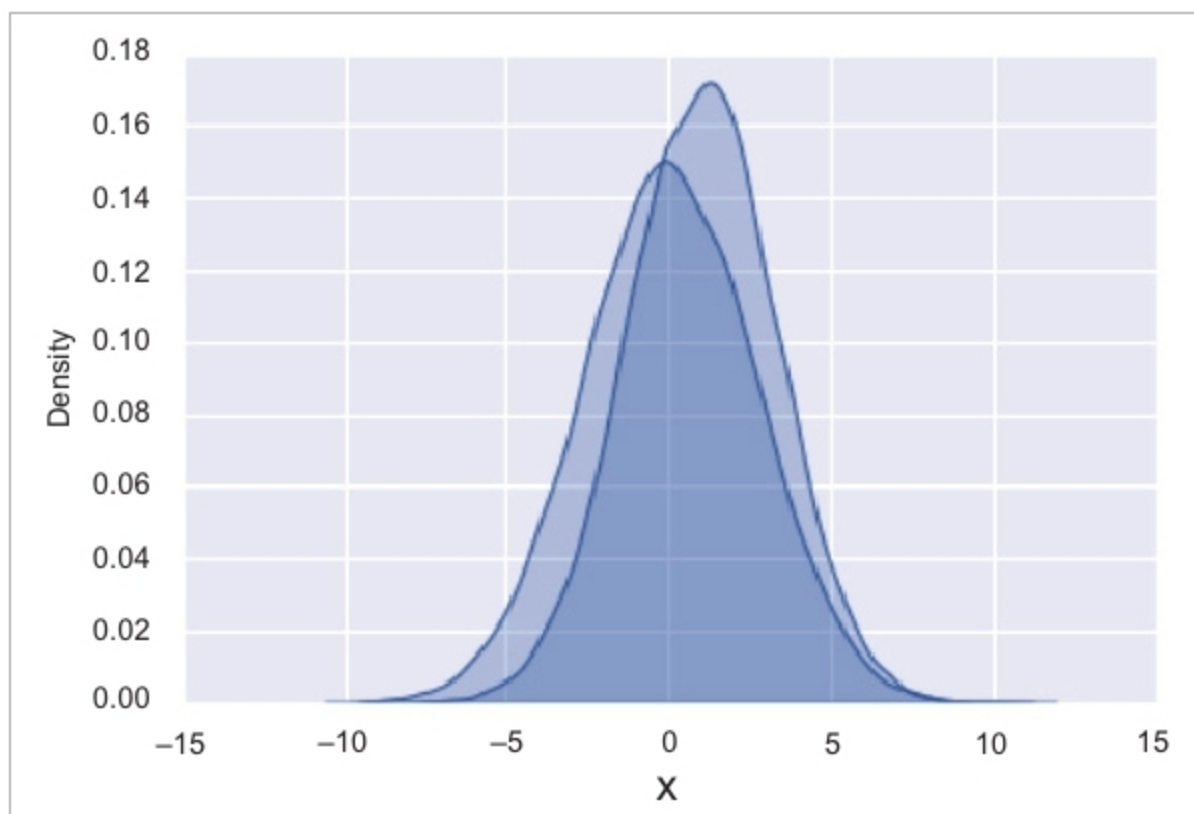


Figure 3: A kernel density estimate (KDE) plot using Seaborn

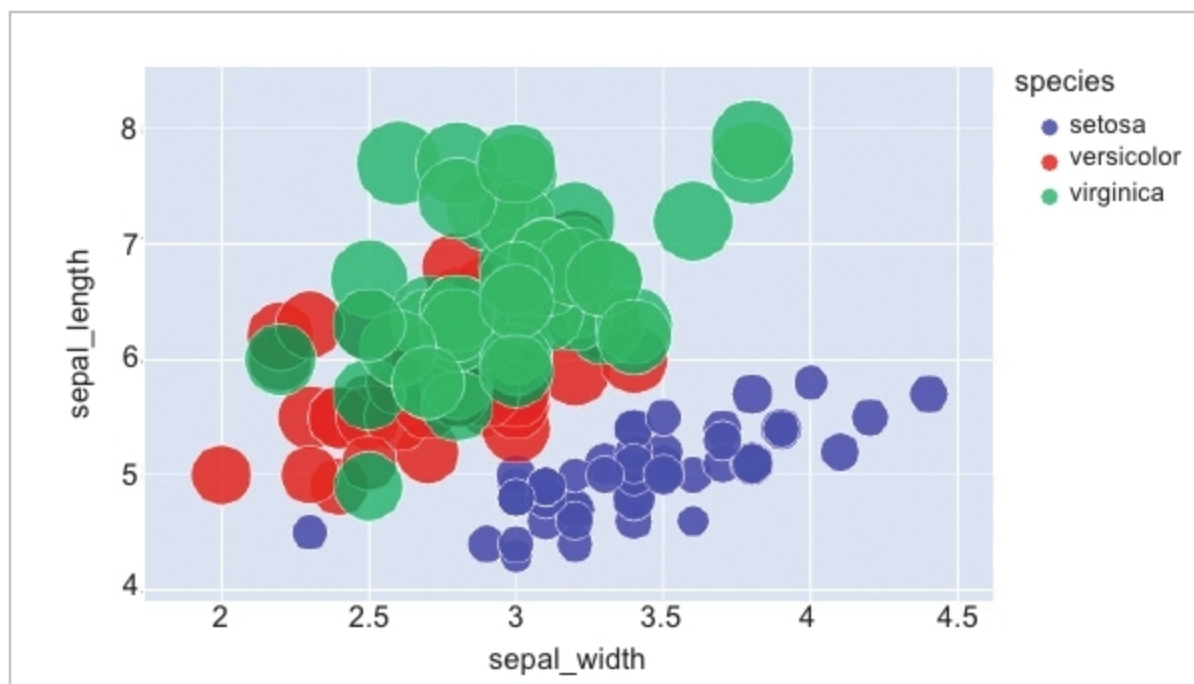


Figure 4: Bubble chart using Plotly

```
import plotly.express as px
iris_dataframe = px.data.iris()
fig = px.scatter(iris_dataframe,
x="sepal_width", y="sepal_length",
color="species",
size='petal_length',
hover_data=['sepal_length'], size_
max=35)
fig.show()
```

The output is shown in Figure 4.

Plotly is a fantastic tool for quick interactive plotting. The code given below shows its power. A 30 line

code gives a clear insight into how COVID-19 spread across countries since its inception. You can experience the interactive part of it at <https://www.kaggle.com/dibyendubanerjee/covid19-spread-visualization>.

```
import plotly.express as px
import pandas as pd
import plotly.graph_objects as go
import numpy as np
url='https://raw.githubusercontent.com/CSSEGISandData/COVID-19/master/
csse_covid_19_data/csse_covid_19_time_
series/time_series_covid19_confirmed_
```

```
global.csv'
```

```
def read_file(url):
    df = pd.read_csv(url)
    return df
```

```
def filter_specific_country(df, selected_
countries):
    df1=df[df['Country/Region'].
isin(selected_countries) ]
    countrywise_grouped_df = df1.
groupby(df['Country/Region']).sum().
drop(['Lat','Long'], axis=1)
    countrywise_grouped_df
    return countrywise_grouped_df
```

```
def transpose_and_reformat_data(df):
    df_t=df.transpose()
    df_t.reset_index(inplace=True)
    df_t.rename(columns={'Country/
Region':'Index_Col', 'index':'Dates'},
inplace=True)
    return df_t
```

```
confirmed_dataset = read_file(url)
selected_countries=['India','China','It
aly','Spain','France','Australia','Germ
any','Japan','Korea, South','Pakistan',
'Russia','United Ki
ngdom','Canada','Iran','Brazil','Singap
ore','South Africa','US']
ds=filter_specific_country(confirmed_
dataset,selected_countries)
data=transpose_and_reformat_data(ds).
melt(id_vars=["Dates"], var_
name="Country", value_name="Confirmed_
Count")
#plot_title="Global Spread of Covid-19:
(Selected Countries)"
plot_title='Visualizing the spread of
Novel Coronavirus COVID-19 (2019-nCoV)
- Created by Dibyendu Banerjee'
fig = px.bar(data, y="Country",
x="Confirmed_Count", color="Country",
animation_frame="Dates", range_
x=[1,14000000], orientation='h' )
fig.update_layout(title=plot_
title,yaxis_title='Countries', xaxis_
tickangle=90, font=dict(family="Arial",
size=10,color="#7f7f7f"))
fig.show()
```